

Model Development Document

Complaint → Regulation Classifier (CMPL-REG-24)
First line of defense — Model Development

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LLM: nim (meta/llama-3.1-8b-instruct)
Data: CFPB Consumer Complaint Database (real, redacted narratives)
Dataset: 4000 rows · regulatory rate 0.9663
Repo: reg-agents · model id: CMPL-REG-24

1 • Executive summary & methodology

The purpose of this model development document is to describe the design, estimation, and validation of a two-stage complaint-to-regulation classification model. This model is intended to classify consumer complaints received by the Consumer Financial Protection Bureau (CFPB) into one of several regulatory categories. The materiality of this model lies in its ability to provide accurate and timely classification of complaints, which is critical for regulatory oversight and enforcement.

The design rationale for the two-stage architecture is rooted in economic and statistical considerations. The first stage, comprising two TF-IDF classifiers (logistic regression and XGBoost), serves as a high-recall gate. This stage is designed to be computationally efficient and to provide a high recall rate, thereby capturing as many complaints as possible. The second stage, utilizing a retrieval-augmented generation pipeline (rag_llm), serves as a more expensive and accurate labeler. This stage is designed to provide a more precise classification of complaints that have been identified as potentially belonging to a specific regulatory category.

The estimation and model-selection protocol involved a bake-off between two TF-IDF classifiers, logistic regression and XGBoost. The champion model, logistic regression, achieved a validation precision-recall area under the curve (PR-AUC) of 0.9913 and a threshold of 0.788. The XGBoost model achieved a validation PR-AUC of 0.9901 and a threshold of 0.844. The confusion matrix for the champion model revealed a true positive rate of 0.9373 and a false positive rate of 0.0718. The second stage, utilizing the rag_llm pipeline, achieved an accuracy of 0.3478 and a family accuracy of 0.5391.

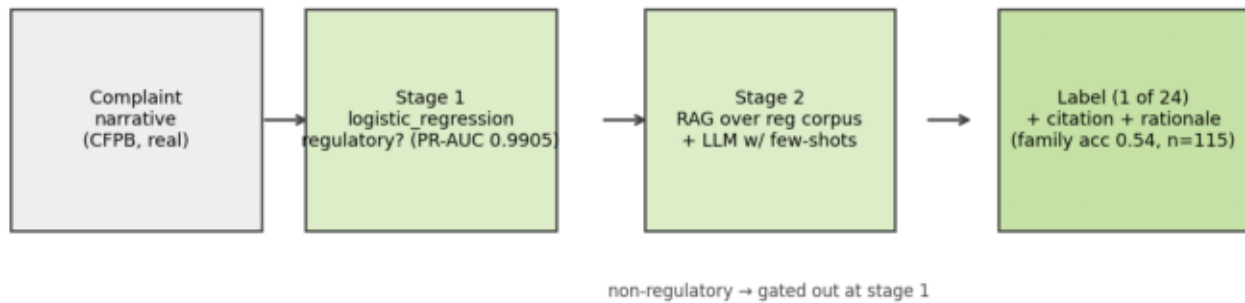
The discrimination of the model is evaluated using both PR-AUC and ROC-AUC metrics. The PR-AUC metric is more relevant in this case due to the class imbalance, with approximately 94% of complaints belonging to the positive class. The champion model achieved a PR-AUC of 0.9913, indicating a high degree of discrimination between the positive and negative classes.

One known weakness of the model is the reliance on weak labels, which may lead to biased or inaccurate classifications. Additionally, the model's performance on certain regulatory categories, such as FCRA_ACCURACY and TILA_ORIGINATION, is relatively poor, with recall rates of 0.2 and 0.0, respectively.

The monitoring and guardrail design is an integral part of the model, rather than an afterthought. The model is designed to be continuously monitored and updated to ensure its accuracy and fairness. The monitoring protocol includes regular evaluation of the model's performance on a holdout set, as well as evaluation of its fairness and bias. The guardrail design includes mechanisms to detect and prevent biased or inaccurate classifications, such as the use of weak labels and the implementation of fairness metrics.

In conclusion, the two-stage complaint-to-regulation classification model is designed to provide accurate and timely classification of consumer complaints received by the CFPB. The model's design and estimation protocol are rooted in economic and statistical considerations, and its performance is evaluated using both PR-AUC and ROC-AUC metrics. While the model has several strengths, including its high recall rate and discrimination, it also has several weaknesses, including its reliance on weak labels and poor performance on certain regulatory categories. The monitoring and guardrail design are integral parts of the model, ensuring its accuracy and fairness.

2 · Architecture



Two-stage pipeline: binary gate, then RAG+LLM labeling with citations.

2 · Architecture (detail)

Stage 1 (binary gate): TF-IDF (1-2 grams, 30k features) into a logistic-regression vs XGBoost bake-off; champion selected on PR-AUC. Upgrade path: fine-tuned BERT-class encoder (NeMo Framework) served via Triton, same interface.

Stage 2 (multi-class): retrieval-augmented generation over the regulation/policy corpus (NeMo Retriever embeddings, FAISS locally / cuVS-Milvus on GPU) + LLM reasoning (NIM Llama-3.1-8B) with 8 curated few-shot examples and the 24-category taxonomy in-prompt. Output is strict JSON: label, confidence, rationale, and the cited source document, which the UI renders alongside the excerpt. A keyword scorer provides a deterministic no-LLM fallback; stage-1 gates non-regulatory complaints so the LLM is only invoked when needed.

Deployment: complaint MCP server (tools: `classify_complaint`, `sample_complaints`, `get_model_metrics`) + Complaint A2A agent; wired into the Streamlit UI and observable via the existing Prometheus/Grafana and OpenTelemetry stack.

3 · Data & curation

Source: real, redacted consumer complaint narratives from the public CFPB Consumer Complaint Database (4000 curated rows; 200 reserved as a 5% scoring holdout for the batch-ingestion layer before any modeling split, then 3040 train / 380 validation / 380 test; regulatory rate 0.9663).

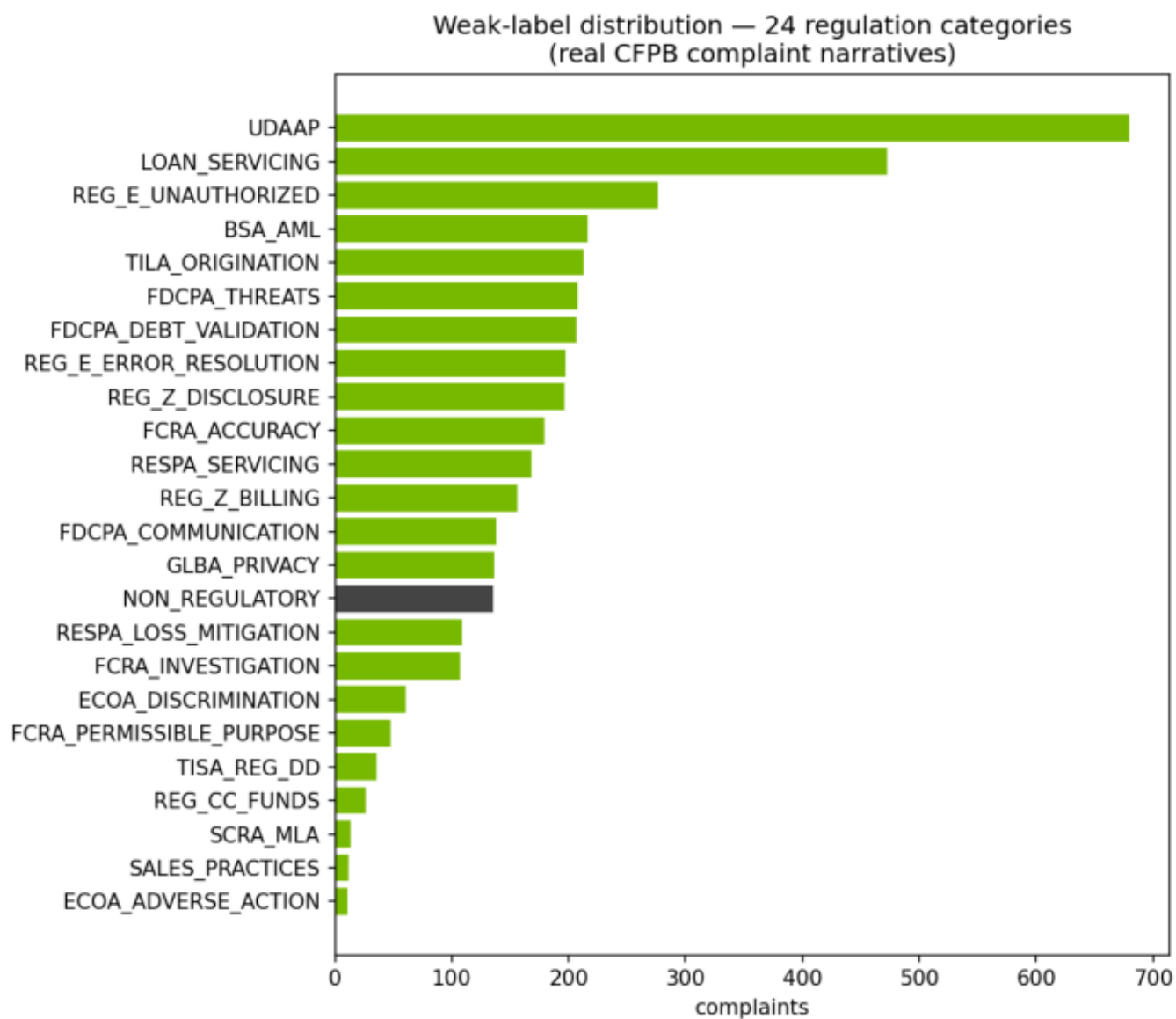
Curation (scripts/fetch_cfpb_complaints.py) mirrors NVIDIA NeMo Data Curator stages: length filtering (ScoreFilter), exact deduplication (ExactDuplicates), near-deduplication (FuzzyDuplicates/MinHash analog), PII verification (CFPB pre-masks PII as XXXX; PiiModifier analog), and per-issue balanced sampling. At corpus scale each stage maps directly to the GPU-accelerated Curator module.

Ground truth is WEAK SUPERVISION: labels derive from the CFPB product/issue taxonomy plus narrative keyword rules across 24 regulation categories (ECOA, FCRA, FDCPA, Reg E, Reg Z, RESPA, UDAP, sales practices, ...). This is the standard bootstrap before human-adjudicated labels exist and is flagged as a limitation in the validation report.

3 · Dataset summary

property	value
source	CFPB Consumer Complaint Database (public, PII-redacted)
curated rows	4,000
scoring holdout (reserved first)	200 (5%, stratified — fed only through the ingestion layer)
train / val / test split	3,040 / 380 / 380 (stratified 80/10/10 of the remaining 95%)
regulatory rate	96.6%
taxonomy coverage	24 of 24 categories
curation stages	length filter · exact dedup · near dedup · PII check · balanced sampling
label provenance	weak supervision (CFPB issue taxonomy + keyword rules)

3 • Label distribution



Weak-label distribution over the 24 regulation categories.

3.1 · Regulation taxonomy (with dataset support)

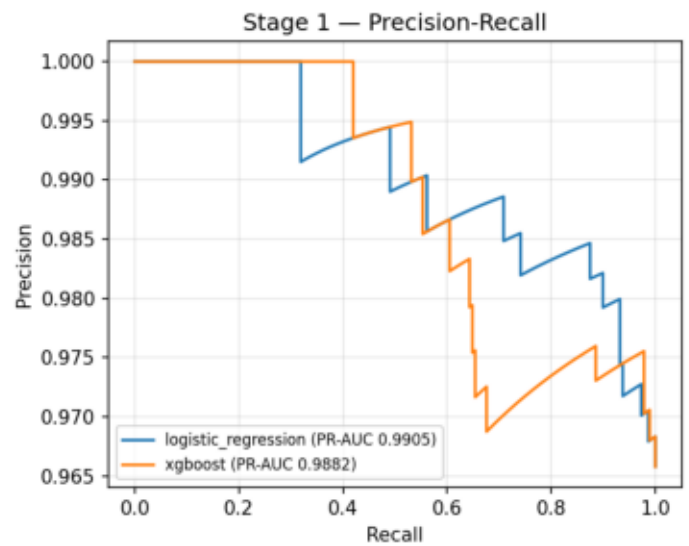
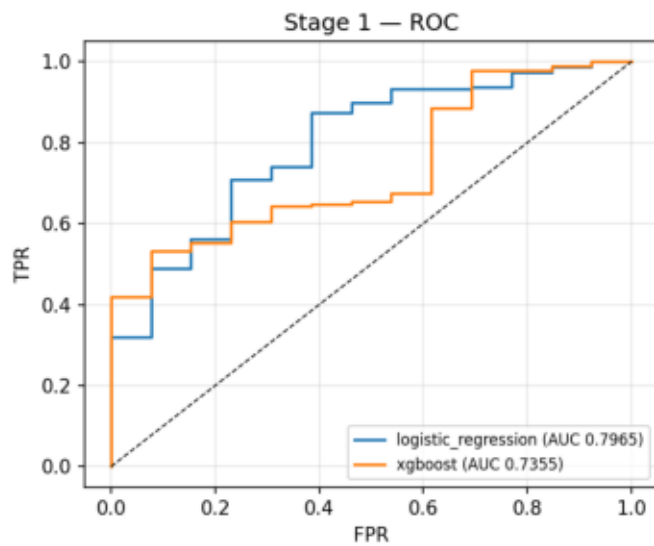
code	regulation / category	n in dataset
FCRA_ACCURACY	FCRA — Accuracy of Reported Information	179
FCRA_INVESTIGATION	FCRA — Reinvestigation of Disputes	107
FCRA_PERMISSIBLE_PURPOSE	FCRA — Permissible Purpose / Improper Use	48
FDCPA_DEBT_VALIDATION	FDCPA — Debt Validation / Not Owed	207
FDCPA_COMMUNICATION	FDCPA — Communication Tactics	138
FDCPA_THREATS	FDCPA — False Statements or Threats	208
REG_E_UNAUTHORIZED	Reg E / EFTA — Unauthorized Transfers	276
REG_E_ERROR_RESOLUTION	Reg E / EFTA — Error Resolution	197
REG_Z_BILLING	Reg Z / FCBA — Billing Error Disputes	156
REG_Z_DISCLOSURE	Reg Z / TILA — Fees, Interest & Disclosures	196
TILA_ORIGINATION	TILA — Credit Origination & Underwriting Disclosures	213
ECOA_DISCRIMINATION	ECOA / Reg B — Credit Discrimination	61
ECOA_ADVERSE_ACTION	ECOA / Reg B — Adverse Action Notices	11
RESPA_SERVICING	RESPA — Mortgage Servicing & Escrow	168
RESPA_LOSS_MITIGATION	RESPA — Loss Mitigation & Foreclosure	109
TISA_REG_DD	TISA / Reg DD — Deposit Account Disclosures	36
REG_CC_FUNDS	Reg CC — Funds Availability	26
BSA_AML	BSA/AML — Account Freezes & Closures	216
GLBA_PRIVACY	GLBA / Privacy — Information Sharing & Safeguards	136
UDAAP	UDAAP — Unfair, Deceptive, or Abusive Acts or Practices	680
SALES_PRACTICES	Sales Practices — Unauthorized Accounts/Products	12
SCRA_MLA	SCRA / MLA — Servicemember Protection	13
LOAN_SERVICING	Loan Servicing (Auto/Student/Personal) — UDAAP & S	472
NON_REGULATORY	Non-Regulatory — General Service	135

4 • Stage-1 bake-off leaderboard

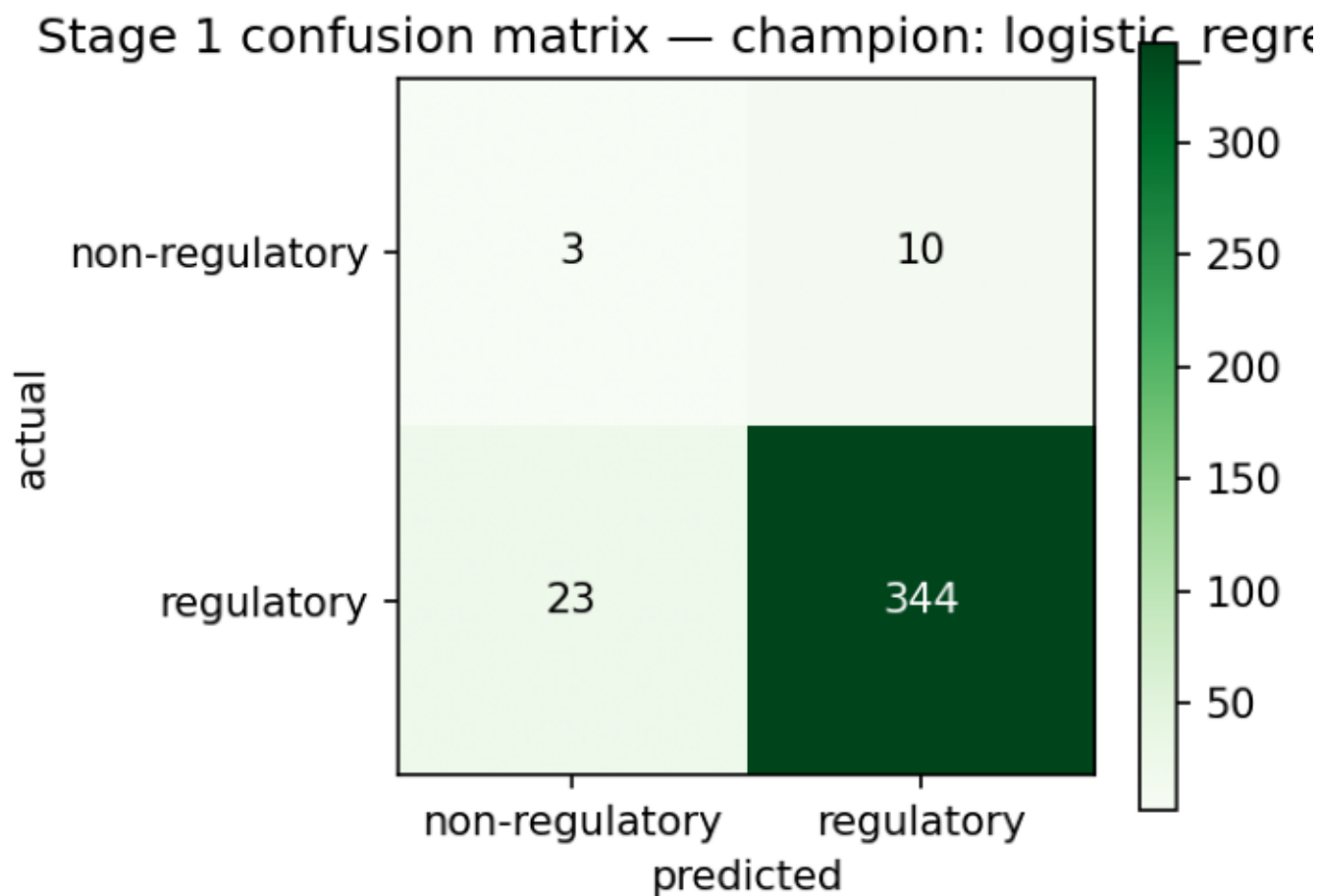
model	val_pr_auc	threshold	pr_auc	roc_auc	f1	precision	recall	accuracy
logistic_regress	0.9913	0.788	0.9905	0.7965	0.9542	0.9718	0.9373	0.9132
xgboost	0.9901	0.844	0.9882	0.7355	0.9453	0.974	0.9183	0.8974

Champion: logistic_regression - selected on validation PR-AUC; other columns are one-shot test metrics.

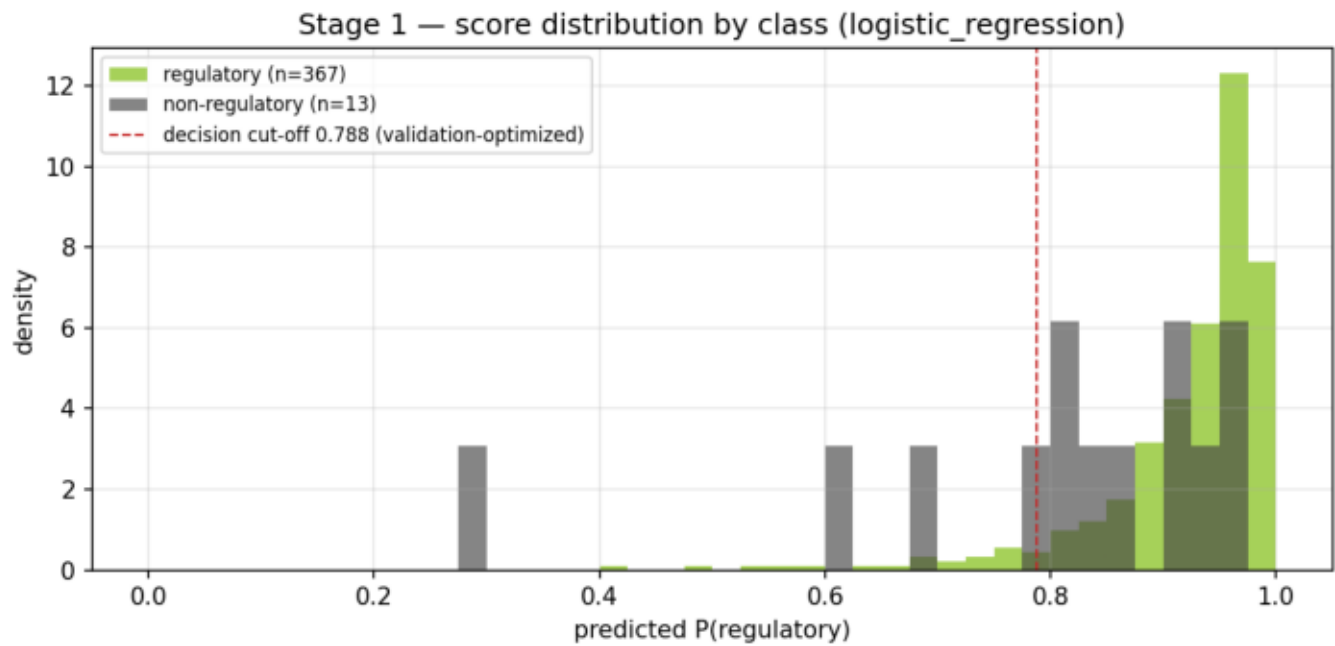
4 • Stage-1 ROC / PR curves



4 • Stage-1 confusion matrix

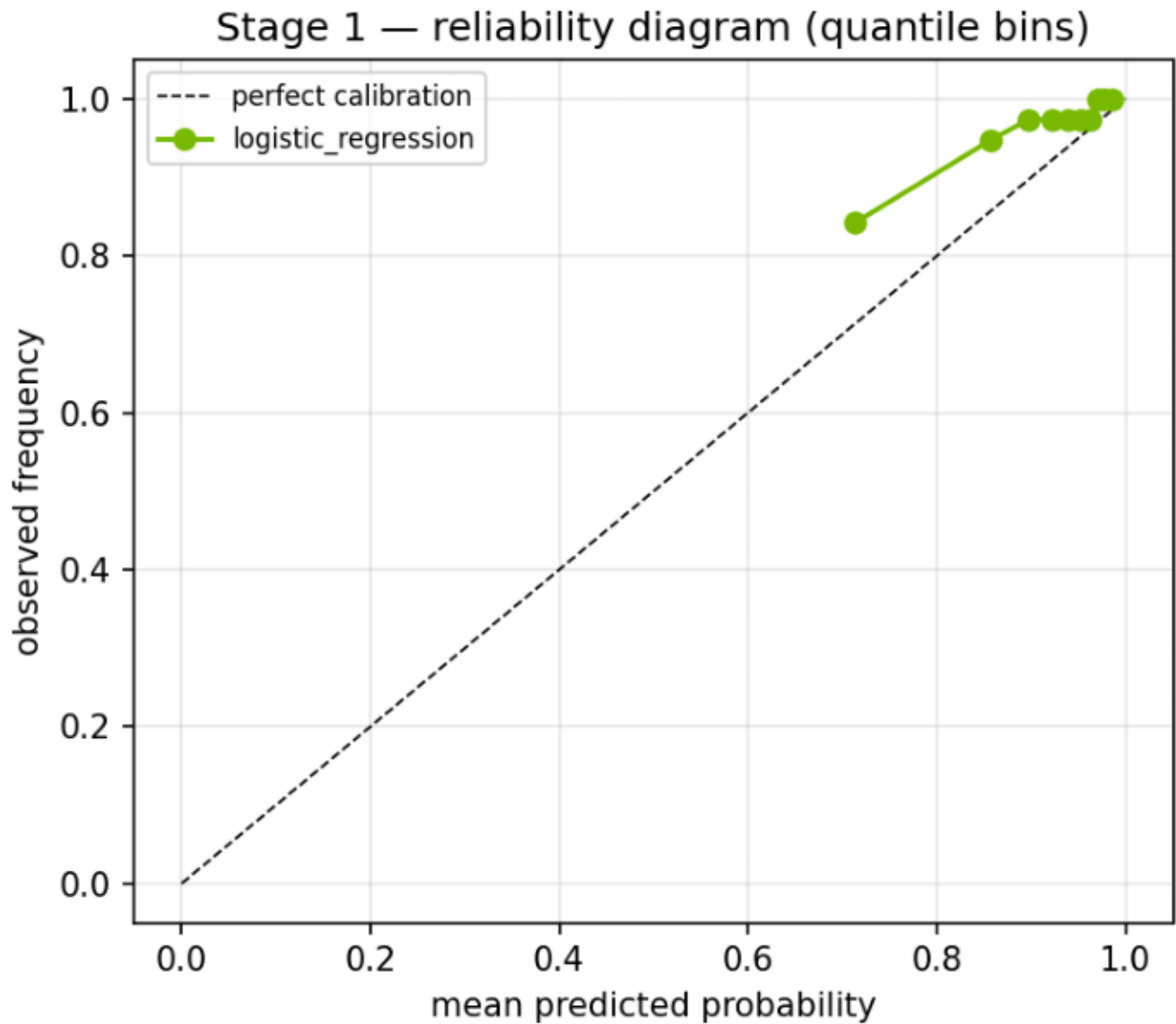


4.1 · Score separation



Class separation at the validation-optimized cut-off 0.788 (default 0.5 not used).

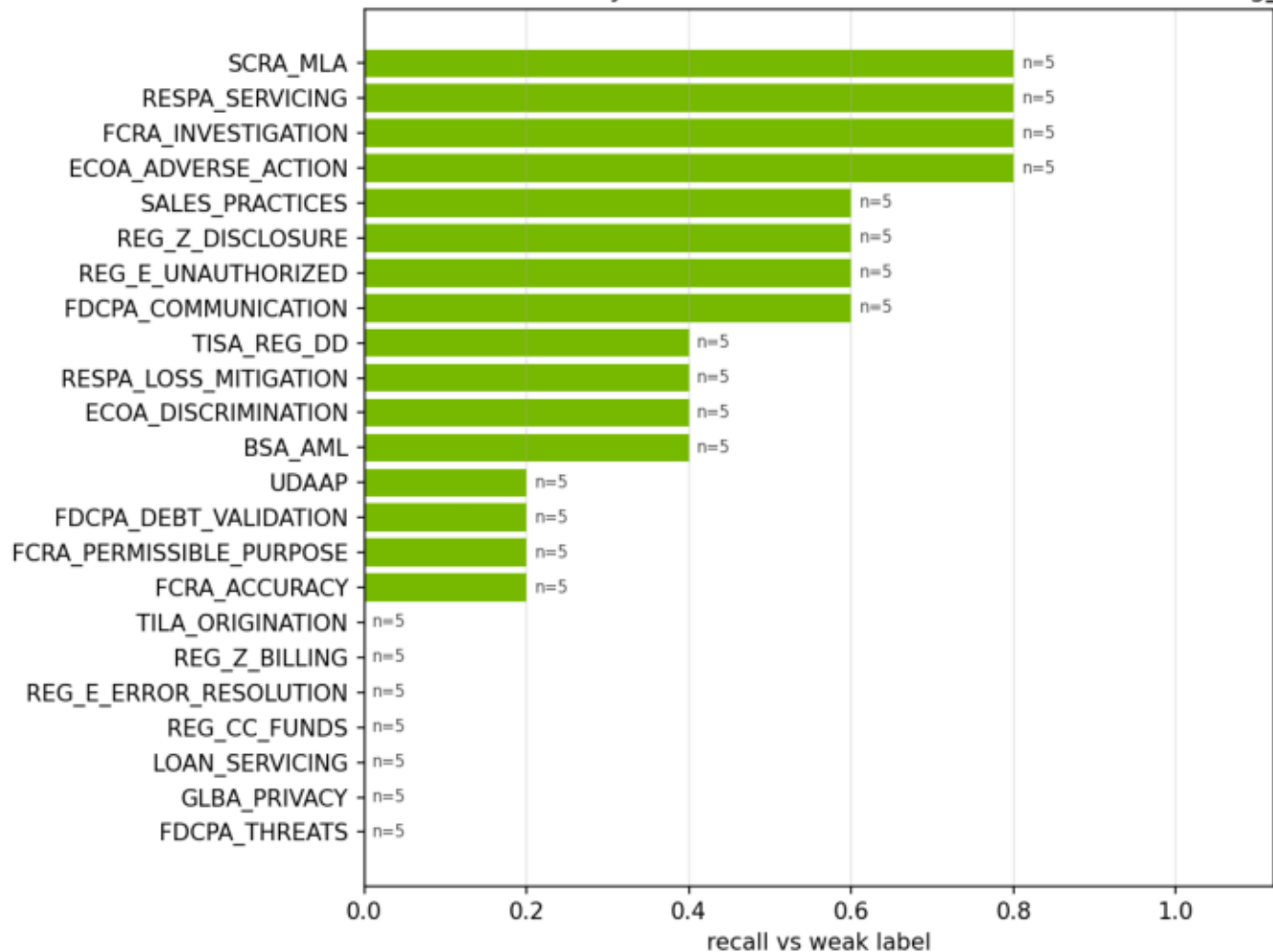
4.1 · Calibration (reliability diagram)



Whether predicted probabilities can be read as probabilities.

5 · Stage-2 per-category recall

Stage 2 (RAG + LLM) — per-category recall
exact acc 0.35 · family acc 0.54 · macro-F1 0.29 · n=115 · mode=rag_ll



Exact acc 0.3478 · family acc 0.5391 · macro-F1 0.2928 · n=115.

5 • Stage-2 per-category detail

category	support	recall
BSA_AML	5	0.4
ECOA_ADVERSE_ACTION	5	0.8
ECOA_DISCRIMINATION	5	0.4
FCRA_ACCURACY	5	0.2
FCRA_INVESTIGATION	5	0.8
FCRA_PERMISSIBLE_PURPOSE	5	0.2
FDCPA_COMMUNICATION	5	0.6
FDCPA_DEBT_VALIDATION	5	0.2
FDCPA_THREATS	5	0.0
GLBA_PRIVACY	5	0.0
LOAN_SERVICING	5	0.0
REG_CC_FUNDS	5	0.0
REG_E_ERROR_RESOLUTION	5	0.0
REG_E_UNAUTHORIZED	5	0.6
REG_Z_BILLING	5	0.0
REG_Z_DISCLOSURE	5	0.6
RESPA_LOSS_MITIGATION	5	0.4
RESPA_SERVICING	5	0.8
SALES_PRACTICES	5	0.6
SCRA_MLA	5	0.8
TILA_ORIGINATION	5	0.0
TISA_REG_DD	5	0.4
UDAAP	5	0.2

6 · Guardrail inventory

guardrail	mechanism	surfaced as
Taxonomy whitelist	LLM label must be one of the 24 codes; else re	complaint_classifications_total{label}
Strict-JSON parsing	malformed LLM output -> deterministic keyword	mode=fallback counter (alertable spike)
Stage-1 gating	non-regulatory volume never reaches the l	cost + attack-surface control
Retrieval grounding	answer must cite a retrieved corpus exce	citation attached to every prediction
Provider portability	OpenA <-> NIM via config; keyword mode if b	mode label per classification

Sign-off

Prepared and reviewed under the bank's model risk management framework (SR 11-7 / OCC 2011-12).

Author, development document: Senior Model Developer & Quantitative Lead — PhD (Econometrics), 20 years in model development across credit, capital planning, fraud, and NLP.

Author, validation report: Senior Independent Model Validator — PhD (Econometrics), 20 years in second-line validation; independent of the development team per SR 11-7 organizational-independence requirements.